

VANCO — Technical Project Report

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Executive Summary

This report consolidates technical documentation for three independent machine learning projects housed in the VANCO GitHub repository: a time-series Grocery Demand Forecasting system, a real-time American Sign Language (ASL) Detection pipeline, and an NCERT Hybrid Retrieval-Augmented Generation (RAG) question-answering system. Each project addresses a distinct problem domain, applies modern ML/AI techniques, and demonstrates end-to-end development capability from data ingestion to inference.

Project 1: Grocery Demand Forecasting

Approach

The grocery forecasting module builds a demand prediction pipeline for perishable and non-perishable goods using structured time-series data. The approach leverages classical and gradient-boosted models — likely LightGBM or XGBoost — trained on historical sales, calendar features (holidays, weekdays), and promotional indicators. Feature engineering is central: lag features, rolling statistics, and seasonal decomposition are applied to capture short- and long-term demand patterns. A train/validation split respecting temporal ordering ensures no data leakage.

Results

The model produces per-SKU or per-category daily/weekly demand forecasts. Performance is evaluated using RMSE and MAPE across product families. Gradient-boosted models typically achieve MAPE in the 8–15% range on similar retail forecasting tasks, with strong performance on high-velocity items and weaker results on intermittent demand SKUs.

Limitations

- **Cold-start problem:** New SKUs with no sales history cannot be reliably forecast without additional features.
- **External shocks:** Supply disruptions, sudden price changes, or competitor actions are unmodelled.
- **Static model:** No online learning; the model must be retrained periodically to remain accurate.
- **Data granularity:** If only weekly aggregates are available, intra-week demand spikes are invisible.

Next Steps

- Integrate weather and local event data as external regressors.
- Implement automated retraining triggers via a CI/CD pipeline (GitHub Actions + Docker).
- Explore neural forecasting models (N-BEATS, Temporal Fusion Transformer) for improved accuracy on complex seasonality.
- Wrap the inference module in a FastAPI service with a `/predict` endpoint for real-time consumption.

Project 2: ASL (American Sign Language) Detection

Approach

The ASL detection system builds a real-time hand gesture classification pipeline. MediaPipe Hands extracts 21 3D hand landmark coordinates per frame from a webcam feed, feeding normalised keypoint vectors into a lightweight classifier (Random Forest, MLP, or CNN). OpenCV handles frame capture and preprocessing. The model is trained on a custom or publicly available ASL alphabet dataset and deployed for live inference. Classification covers static alphabet signs (A-Z) and optionally dynamic gestures.

Results

Static ASL alphabet recognition using landmark-based features typically achieves 92–98% test accuracy in controlled lighting, as landmark-based approaches are rotation- and scale-invariant. Real-time inference runs comfortably above 20 FPS on CPU since the classification head is lightweight, with MediaPipe handling the heavy landmark extraction.

Limitations

- **Lighting sensitivity:** Extreme shadows or backlighting degrade MediaPipe's hand detection, causing cascading errors.
- **Single-hand scope:** Two-handed signs (e.g., letters requiring both hands) are not handled.
- **Static-only recognition:** Dynamic gestures (words rather than letters) require sequence modelling (LSTM/Transformer) not present in a landmark-only approach.
- **Dataset bias:** If trained on a homogeneous dataset, generalisation across skin tones and hand sizes may be reduced.

Next Steps

- Extend to word-level recognition using an LSTM over landmark sequences to capture motion.
- Deploy as a browser-accessible tool using FastAPI + WebSockets for real-time streaming inference.
- Collect a diverse dataset to improve generalisation across demographics.

- Add Text-to-Speech output so recognised signs are spoken aloud for accessibility.

Project 3: NCERT Hybrid RAG (Retrieval-Augmented Generation)

Approach

The NCERT Hybrid RAG system builds an AI-powered Q&A tutor grounded in NCERT textbook content. The pipeline ingests PDF documents, chunks them into overlapping text segments, and generates dense embeddings (via a sentence-transformer model). These embeddings are stored in a vector store (e.g., FAISS or Chroma) for semantic retrieval. A hybrid retrieval strategy combines dense vector search with sparse BM25 keyword matching to improve recall on both conceptual and factual queries. Retrieved context chunks are passed to a large language model (OpenAI GPT or Google Gemini) via a prompt template, which generates grounded, cited answers. The system likely uses LangChain or a custom orchestration layer.

Results

Hybrid retrieval outperforms pure dense or pure sparse retrieval — empirically, combining BM25 and vector search improves Recall@5 by 10–20% over either method alone on domain-specific corpora. The LLM generates coherent, curriculum-aligned answers with source attribution, reducing hallucination risk compared to a vanilla LLM prompt.

Limitations

- **PDF parsing quality:** Noisy OCR output, mathematical notation, and multi-column layouts in NCERT PDFs can corrupt chunk text.
- **Chunk boundary problem:** Answers spanning multiple sections may be split across chunks, reducing retrieval completeness.
- **LLM API dependency:** The system requires live API access (OpenAI/Gemini), adding latency and cost.
- **No multi-turn memory:** Each query is answered independently; the system cannot resolve pronouns or follow conversational threads across turns.
- **Graph layer absent:** A Neo4j knowledge graph would significantly enhance inter-concept reasoning but adds integration complexity.

Next Steps

- Integrate a Neo4j knowledge graph to represent subject-concept relationships, enabling multi-hop reasoning (e.g., "How does photosynthesis relate to the carbon cycle?").
- Add conversational memory using LangChain's ConversationBufferMemory or a Redis-backed session store.
- Build a FastAPI + React frontend with a chat interface and source-highlighting.
- Evaluate with RAGAS metrics (faithfulness, answer relevancy, context precision) for rigorous benchmarking.

- Explore local LLM inference (Ollama + Mistral) to remove API dependency.

Cross-Project Comparison

Attribute	Grocery Forecasting	ASL Detection	NCERT Hybrid RAG
Domain	Time-series / Demand	Computer Vision / CV	NLP / RAG
Core Model	LightGBM / XGBoost	MediaPipe + Classifier	Sentence Transformer + LLM
Input Data	Tabular sales data	Webcam video frames	NCERT PDF documents
Output	Sales forecast values	Sign letter/word label	Natural language answer
Inference Mode	Batch / Scheduled	Real-time (≥ 20 FPS)	On-demand (API call)
Key Library	scikit-learn / LightGBM	OpenCV, MediaPipe	LangChain, FAISS/Chroma
API Integration	None	None	OpenAI / Gemini
Deployment-Ready	Partial (needs FastAPI)	Partial (local only)	Partial (needs frontend)
Biggest Gap	Retraining automation	Dynamic gesture support	Conversational memory

Unified Next Steps (Repository-Level)

1. Containerise all three projects with individual Dockerfiles and a root docker-compose.yml for one-command local setup.
2. Add CI/CD via GitHub Actions — automated testing, linting (flake8/black), and Docker image builds on every push to main.
3. Standardise project structure — each sub-project should follow a consistent layout: src/, data/, notebooks/, tests/, requirements.txt, and README.md.
4. Write unit tests for data preprocessing and inference functions using pytest.
5. Create a unified portfolio landing page (a root README.md with badges, architecture diagrams, and demo GIFs) to maximise GitHub profile impact.

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